LEVEL UP ACADEMY

Class 9 | CBSE / GSEB | Science

Unit Test: Motion

Chapter / Topic: Motion (topic-based)

Name: _____ Roll No.: _____ Date: _____

Time: 50 minutes **Max. Marks: 25**

General instructions:

1. All questions are compulsory. Internal choice is given in some questions.
2. Section A: 1 mark each. Section B: 2 marks each. Section C: 3 marks each. Section D: 5 marks. Section E: case-based, 4 marks.
3. Show the formula, substitution and unit in every numerical.

SECTION A

6 marks

Choose the correct option.

Q1. The SI unit of acceleration is: **[1]**

(a) m/s (b) m s
(c) s/m (d) m/s^2

Q2. The slope of a velocity–time graph gives: **[1]**

(a) speed (b) distance
(c) acceleration (d) displacement

Q3. A body moving along a circular path at constant speed has: **[1]**

(a) constant velocity (b) zero acceleration
(c) changing velocity (d) zero velocity

Q4. A car goes from rest to 20 m/s in 4 s. Its acceleration is: **[1]**

(a) 0.2 m/s^2 (b) 5 m/s^2
(c) 16 m/s^2 (d) 80 m/s^2

Q5. Which of these can be zero even when a body has moved? **[1]**

(a) Distance (b) Speed
(c) Time taken (d) Displacement

Q6. **Assertion (A):** A body moving with uniform velocity has zero acceleration.
Reason (R): Acceleration is the rate of change of velocity. **[1]**

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

SECTION B

4 marks

Q7. Give two differences between distance and displacement. **[2]**

OR

Give two differences between speed and velocity.

Q8. A cyclist completes one round of a circular track of radius 21 m in 44 s. Find the cyclist's speed. (Take $\pi = 22/7$.) **[2]**

SECTION C

6 marks

Q9. A train speeds up uniformly from 10 m/s to 30 m/s in 10 s. **[3]**

(i) Find its acceleration. **[1]**

(ii) Find the distance covered in these 10 s. **[1]**

(iii) What will its velocity be after another 5 s at the same acceleration? **[1]**

Q10. Derive the equation $v = u + at$ using a velocity–time graph. **[3]**

OR

A car moving at 72 km/h is braked uniformly and stops in 4 s. Find the retardation and the stopping distance.

SECTION D

5 marks

Q11. Answer the following: **[5]**

(i) Derive the equation $s = ut + \frac{1}{2}at^2$ using a velocity–time graph. **[3]**

(ii) A ball rolls down a slope from rest with uniform acceleration and covers 9 m in 3 s. Find its acceleration. **[2]**

SECTION E

4 marks

Read the passage and answer the questions that follow.

Q12. A student recorded the distance travelled by a scooter on a straight road: at t = 0, 2, 4 and 6 s, the distance from the start was 0, 10, 20 and 30 m. **[4]**

(i) What type of motion does the scooter have? **[1]**

(ii) Find the speed of the scooter. **[1]**

(iii) What will the distance be after 10 s at the same speed? What does the distance–time graph look like? **[2]**

~ All the best! ~